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Assignment 1: observations on infection.txt

Running the C code found in the translated infection.txt on a VM running Linux Mint caused the computer to, essentially, lose all functionality, with running programs freezing and preventing the startup of any new programs. Attempting to open new programs so created prompted an error for "failed to fork (resource temporarily unavailable)". This even prevented me from logging out or even shutting the machine down via the taskbar menu.

Running the same code on a Windows 11 machine yielded delayed but equally severe results. For many minutes after execution, there were no noticeable effects. The only obvious symptom at this point was that Task Manager appeared to be going completely haywire and would show a garbled user interface before locking up and causing severe lag. After about half an hour, however, similar symptoms to the Mint machine suddenly appeared, with programs freezing and new ones refusing to launch. This forced a restart of the computer.

Given these observations, I believe this code is a 'fork bomb'; it repeatedly opens copies of itself and floods the system's memory. This may be useful for attackers as a crude form of denial-of-service attack, causing a machine to be effectively crippled by preventing process execution. This likely leaves the system in a vulnerable state where it may be susceptible to attack, or it may also force the system to be restarted, allowing a brief window without protection. One possible method to counter such an attack would be to detect how much memory a program is using and to forcibly restrict that amount or kill the program if it consumes too much.